

ECON 101H, Introduction to Economics

Recitation 6: Taxation

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Demand and supply are:

$$Q_D = 120 - P, \quad Q_S = -240 + 3P.$$

(a) Equilibrium and Total Surplus

Set demand equal to supply:

$$120 - P = -240 + 3P \Rightarrow 4P = 360 \Rightarrow P^* = 90.$$

Substitute into either equation:

$$Q^* = 120 - 90 = 30.$$

The choke prices are:

$$P_D = 120, \quad P_S = 80.$$

Total surplus is the area of the triangle:

$$TS = \frac{1}{2}(120 - 80)(30) = \frac{1}{2}(40)(30) = 600.$$

(b) Policy and Entry Effects

Allowing calculators on exams increases demand. Entry of a new producer increases supply.

$\Rightarrow$ Quantity increases, price is ambiguous.

(c) \$20 Tax on Buyers

Rewrite demand as:

$$P = 120 - Q.$$

With a \$20 tax:

$$P = 100 - Q.$$

New demand:

$$Q = 100 - P.$$

Set equal to supply:

$$100 - P = -240 + 3P \Rightarrow 4P = 340 \Rightarrow P = 85.$$

$$Q = 100 - 85 = 15.$$

15 calculators will be purchased.
